Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 31)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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7	
8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows the annual mean concentrations of four pollutants measured at a city-centre monitoring station, with the legal limit for each.

Table 1

Pollutant	Annual mean / µg m^{-3}	Legal limit / µg m^{-3}
Nitrogen dioxide	52	40
PM10 particulates	28	40
PM2.5 particulates	19	20
Sulfur dioxide	8	125

0 1 . 1 Calculate by what percentage the annual mean nitrogen dioxide concentration exceeds the legal limit.

[2 marks]

Answer ☐ %

0 1 . 2 Suggest why the sulfur dioxide concentration is now far below its limit while nitrogen dioxide is not.

[3 marks]

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0 1 . 3 Explain how each of the following reduces emissions.

[3 marks]

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Electrostatic precipitator .....

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Catalytic converter .....

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Low-NOx burner .....

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0 1 . 4 Explain why a reduction in the city's NOx emissions would also reduce summer ground-level ozone.

[3 marks]

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0 1 . 5 The council also plans to monitor air quality using lichens. Explain the advantages and the limitations of using lichens as indicator species.

[4 marks]

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0 2 . 1 Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

A High sand content ☐

B High clay content ☐

C Crumb peds ☐

D High pH ☐

0 2 . 2 Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

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0 2 . 3 Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

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0 3 . 1 Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

A Sulfur dioxide from a coal-fired power station ☐

B Tropospheric ozone ☐

C Carbon monoxide from a car exhaust ☐

D Nitrogen monoxide from a diesel engine ☐

0 3 . 2 Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

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0 4 . 1 Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen ☐

B Oxygen ☐

C Argon ☐

D Carbon dioxide ☐

0 4 . 2 Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

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0 4 . 3 Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

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0 5 . 1 Explain how the control of emission **location** and emission **timing** can reduce the severity of pollution. Use examples.

[6 marks]

Emission location .....

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Emission timing .....

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0 5 . 2 Explain the **polluter pays** principle and suggest why it encourages pollution prevention.

[2 marks]

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0 5 . 3 Explain the **precautionary principle** as applied to the release of a new chemical.

[2 marks]

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0 6 . 1 Complete Table 1 to show one method of controlling each source of railway noise.

[4 marks]

Table 1

Source of railway noise	Control method	
Wheel vibration on the track		
Pantograph turbulence		
Wheel squeal on corners		
Braking squeal		

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0 6 . 2 Explain why road traffic noise affects more people than any other form of noise pollution.

[1 mark]

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0 6 . 3 Describe **three** ways of reducing road traffic noise reaching residents.

[3 marks]

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0 6 . 4 Suggest how noise from pile-driving at sea for wind turbine foundations can be reduced to protect cetaceans.

[2 marks]

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0 7 A student investigated the effect of pH on the germination of cress seeds.

0 7 . 1 Explain why cress is a suitable species for this investigation but the results may not apply directly to natural ecosystems.

[2 marks]

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0 7 . 2 Describe how the student should decide the range of pH values to test.

[2 marks]

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0 7 . 3 Give **three** variables that should be standardised and explain how long the experiment should continue.

[4 marks]

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0 7 . 4 The student wants to compare the effect of nutrient concentration on the growth of duckweed. Explain why leaf area is a better measure of growth than fresh mass.

[2 marks]

10

0 8 . 1 Explain the difference between a negative and a positive feedback mechanism.

[2 marks]

0 8 . 2 Describe **two** negative feedback mechanisms that may reduce global warming.

[4 marks]

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0 8 . 3 Explain how melting permafrost and melting methane hydrate act as positive feedback mechanisms.

[4 marks]

0 8 . 4 Give **three** other positive feedback mechanisms.

[3 marks]

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0 8 . 5 Explain what is meant by a climate **tipping point**.

[2 marks]

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0 9 . 1 Give **three** sources of mercury pollution.

[3 marks]

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0 9 . 2 Describe **three** methods used to control mercury pollution.

[3 marks]

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0 9 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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0 9 . 4 State what is meant by synergism and give an example involving a heavy metal.

[2 marks]

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1 0 . 1 Evaluate the extent to which new technologies could allow fossil fuels and nuclear fission to continue to supply a large proportion of the world's energy sustainably.

[9 marks]

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Extra space

1 0 . 2 State the approximate percentage of natural gas in a reservoir that can be recovered using its own pressure.

[1 mark]

10

Write an essay on one of the following topics.

1 1 . 1 To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?

[25 marks]

OR

1 1 . 2 Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS